JAYADITYA VETSA

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EDUCATION

THE UNIVERSITY OF TEXAS AT AUSTIN, *Austin, TX* Class of 2030
– Bachelor of Science in Computer Science w/ Robotics Honors

ROUSE HIGH SCHOOL, *Leander, TX* Class of 2026 9, 10, 11, 12
GPA: **4/4** unweighted; **5.672** weighted; Class Rank: **5/511** SAT: **1540**; AP CS A (**5**), AP Calc BC (**5**), AP Research (**5**), etc…

SKILLS

- **Languages**: Python, Java, Javascript, TypeScript, C++, Kotlin, Swift/iOS
- **AI & Frameworks**: PyTorch, LlamaIndex, LangChain, LSTMs, OpenCV, Tensorflow, NiBabel, Skimage
- **Tools**: Git/GitHub, Azure, Google Cloud, n8n, Vercel, Railway, Postman, OnRender

EXPERIENCE & RESEARCH

NOVA: Deep Learning for Early Detection of Neurodegenerative Diseases | Independent Research

- Developed a gray matter segmentation model using **NiBabel and skimage** to detect and visualize neurodegenerative disease progression **using MRI/fMRI scans**, focusing on early-stage Parkinson's and paralysis indicators
- Selected for **Round 2 Judging** at the Greater Austin Regional Science and Engineering Fair **(GARSEF)** and **published preprint in IEEE preprint server, TechRxiv**

The Impact of Sentiment Source Credibility on LSTM-Based Stock Price Prediction | Independent Research

- Conducted independent research on the role of **sentiment source credibility in forecasting stock prices** using **LSTM neural networks** and **published preprint in IEEE preprint server, TechRxiv**
- **Built a custom** Python-based prediction **pipeline combining VADER sentiment analysis**, stock data from Yahoo Finance, and multiple news/sentiment sources (real, Reddit, and AI-generated) that showcased that **credibility of sentiment sources significantly impacts model accuracy**

Calimatic Ed Tech | ML Intern

- Developed and deployed a chatbot with **1,000+** successful executions using Python and APIs
- Leveraged advanced tools like **OpenAI API, LangChain, Azure, LlamaIndex** to design scalable AI solutions, and built **5+ automated workflows in n8n**, streamlining lead management, and SEO processes

Single-Frame Action Recognition on AVA Dataset | Independent Project

- Engineered a **computer vision pipeline** utilizing a fine-tuned **EfficientNet-B2** model to classify complex human actions reaching a **mAP of 12.99%**
- Conducted rigorous error analysis on misclassifications, identifying the **structural limitations of spatial-only models**

Leander ISD Survey App | Co-Founder & Lead Engineer

- **Built and published** an iOS survey app to replace **$10k+/yr third-party tools** like Google Forms, saving the district time and money utilizing real-time feedback features to boost student participation, under **beta testing with RHS administrators and district officials, with potential 25,000+** users across LISD

Quantify IQ | Co-Founder & Lead Backend Developer (QuantifyIQ)

- Improved financial literacy for **50+ teens**, featured in the Young Investors Society newsletter (**12,000+ students reached**) includes **Efficient Frontier Model**, **Black-Litterman model**, **Monte Carlo simulations**, and **stress testing**

North South Foundation | Volunteer (150+ hours)

- Tutored **100+ students** in advanced math and raised **$4,000+** to fund scholarships for low-income students in India and helped create asynchronous class material for spelling bee